Gradients of Straight Lines - Mark scheme

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 12

12	$y=\frac{7-5x}{2}$ or $y=\frac{7}{2}-\frac{5}{2}x$ or $y=3.5-2.5x$ or $2y=7-5x$ oe		2	M1 for making y or $2y$ the subject Allow $y=-\frac{5}{2}x+c$ oe
		-2.5		A1 for $-\frac{5}{2}$ or -2.5
				Total 3 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2026 · Question 15

Question	Working	Answer	Mark	Notes
	$\frac{4--6}{-3-2}\left(=\frac{10}{-5}\right)$ oe or $\frac{-6-4}{2--3}\left(=\frac{-10}{5}\right)$		2	M1 method to find gradient. Condone correct gradient embedded in an equation eg $y=-2x+c$ **or** an expression eg $-2x$ **or** for an answer of 2 from $10\div5$ or from $-10\div-5$
	Correct answer scores full marks (unless from obvious incorrect working)	-2		A1 cao an answer of $y=-2x-2$ scores M1 only
				Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 18

18	Gradient of **P** $=-\frac{2}{5}$ or $y=\frac{7-2x}{5}$ oe or $y=-0.4x+\ldots.$ or $\frac{5}{2}x$ or $y=\frac{5}{2}x(+\ldots.)$		2	M1 oe for the given equation rearranged so it is possible to see the gradient or for an equation with gradient of 2.5 oe or $\frac{5}{2}x$ oe
	Correct answer scores full marks (unless from obvious incorrect working)	$\frac{5}{2}$		A1 oe gradient must be stated but isw if seen and then used in an equation SCB1 for the equation of a line with gradient $\frac{5}{2}$ if $\frac{5}{2}$ not seen separately and no other mark awarded
				Total 2 marks

Question 4

Practice question written for this worksheet (not from a past paper)

Line L has equation $y=2x+3$.

Find the equation of the line parallel to L that passes through $(1, -4)$.

Gradient 2: $y=2x+c$

$-4=2+c$, so $c=-6$

Answer: y = 2x - 6

Question 5 *Practice question written for this worksheet (not from a past paper)*

Find the equation of the line perpendicular to $y = \frac{1}{2}x + 1$ that passes through (2, 5

Perpendicular gradient: -2

$5 = -4 + c$, so $c = 9$

Answer: y = -2x + 9

Question 6 *Practice question written for this worksheet (not from a past paper)*

Find the coordinates of the point where the lines $y = 2x + 1$ and $y = -x + 7$ meet.

$2x + 1 = -x + 7$, so $3x = 6$, $x = 2$

$y = 2 \times 2 + 1 = 5$

Answer: (2, 5)

Question 7 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 10*

Q	Working	Answer	Mark	Notes
(a)	eg $2y = -7x(+10)$		2	M1 for $2y = -7x(+10)$ or an answer of $-3.5x$ oe or an answer of 3.5 oe
		-3.5		A1 oe
(b)		(0, 5)	1	B1 cao
				Total 3 marks

Question 8 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2019 · Question 11*

11	e.g. $y = 2 - \frac{1}{2}x$ or $y = 2 - \frac{x}{2}$ or $y = \frac{4-x}{2}$ or gradient of $L_1 = -0.5$ oe	Yes, with correct gradients shown to make -1 when multiplied	3	M1
	e.g. $\frac{9--7}{7--1}$ (=2) or $\frac{-7-9}{-1-7}$ (=2)			M1
				A1 $2 \times -0.5 = -1$ and yes
Alternative scheme for 11				
	e.g. $y = 2 - \frac{1}{2}x$ or $y = 2 - \frac{x}{2}$ or $y = \frac{4-x}{2}$ or gradient of $L_1 = -0.5$ oe	Yes, with correct equation shown to be valid by using the given points	3	M1
	$-7 = 2(-1) + c$ or $9 = 2(7) + c$ ($c = -5$)			M1 dep on M1 for substituting $(-1, -7)$ or $(7, 9)$ into $y = 2x + c$ to find value of c
				A1 Uses the other point in $y = 2x - 5$ to show it is valid and yes

Question	Working	Answer	Mark	Notes

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2026 · Question 13

Q	Working	Answer	Mark	Notes
(a)	eg $4y = 10x$ (+ 5) oe eg $y = 2.5x$ (+ 1.25) or $-10x$ (−5) = $-4y$ oe		2	M1 for $4y = 10x$ (+ 5) oe **or** an answer of $2.5x$ oe **or** an answer of -2.5 oe
	Correct answer scores full marks (unless from obvious incorrect working)	2.5		A1 oe eg $\frac{5}{2}, \frac{10}{4}, \frac{-5}{-2}, \frac{-10}{-4}$
(b)		(0, 1.25)	1	B1 oe eg $\left(0, \frac{5}{4}\right)$
				Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2021 · Question 13

Q	Working	Answer	Mark	Notes
(a)	$\frac{8--4}{5-9}\left(=\frac{12}{-4}\right)$ oe or $\frac{-4-8}{9-5}\left(=\frac{-12}{4}\right)$ oe		2	M1 condone correct gradient embedded in an equation e.g. $y = -3x + c$ or expression e.g. $-3x$ or for an answer of 3
		−3		A1
(b)		$\frac{1}{4}$	1	B1 accept 0.25 or $-\frac{1}{-4}$ oe
				Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 13

Q	Working	Answer	Mark	Notes
	eg (gradient =) $\frac{12--48}{-5-19}\left(=-2.5\right)$ oe		3	M1 for a method to find the gradient
	eg $12 = \text{“}-2.5\text{”} \times -5 + c$ oe $y - 12 = \text{“}-2.5\text{”}(x - -5)$ oe			M1 ft their gradient
	Correct answer scores full marks (unless from obvious incorrect working)	$y = -2.5x - 0.5$		A1 oe eg $y - 12 = -2.5(x + 5)$ **or** $2y + 5x + 1 = 0$
				Total 3 marks

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Question 12 *Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2016 · Question 18*

18	eg. $\frac{3--1}{1--2}\left(=\frac{4}{3}\right)$			M1 for gradient of line AB	M1 for $y=\frac{4}{3}x+\frac{5}{3}$	M2 for sketch of **L** with $(0, \frac{4}{3})$ and $(2, 0)$ marked on axes **AND** (1,3) and (-2 , -1) joined
	$y=\frac{4}{3}-\frac{2}{3}x$ **or** $y=\frac{4-2x}{3}$ **or** $(m=)-\frac{2}{3}$			M1	M1 for $\frac{4}{3}x+\frac{5}{3}=\frac{4}{3}-\frac{2}{3}x$ oe	
		No with reason	3	A1	accept no with $-\frac{2}{3}$ (or $-0.666\ldots$) **and** $\frac{4}{3}$ (or $1.333\ldots$) $-\frac{2}{3}$ and $\frac{4}{3}$ may be seen as coefficients of x in $y=mx+c$ **OR** shows that $3y=4-2x$ and line through AB ($y=\frac{4}{3}x+\frac{5}{3}$) intersect at $x=-\frac{1}{6}$ **or** $y=\frac{13}{9}$ **OR** **L** has a negative gradient (with evidence), line AB has a positive gradient (with evidence) NB: Any value given for a gradient must be correct	
						Total 3 marks

Question 13 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 20*

Q	Working	Answer	Mark	Notes
	Gradient of **M** $=-\frac{1}{4}$ oe		3	M1 for a statement that the gradient of **M** is $-\frac{1}{4}$, may be implied by equation of line with gradient of $-\frac{1}{4}$
	eg $1="-\frac{1}{4}"\times 8+c$ or $y-1="-\frac{1}{4}"(x-8)$			M1 ft a gradient of $\frac{1}{4}$ or -4 for this mark, substitution of the given coordinate to find the equation
	Correct answer scores full marks (unless from obvious incorrect working)	$y=-\frac{1}{4}x+3$		A1 oe $y=-0.25x+3$ must be in the form $y=$
				Total 3 marks

Question 14 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2018 · Question 13*

13	eg $m=\frac{1}{2}$ or $y=\frac{1}{2}x+c$		4	M1	for gradient $=\frac{1}{2}$
	eg $7=\frac{1}{2}\times 4+c$ or $y-7=\frac{1}{2}(x-4)$			M1	for substituting (4,7) into an equation with gradient $=\frac{1}{2}$
	eg $\frac{1}{2}x+5=0$ or $-7=\frac{1}{2}(x-4)$			M1	Inputting $y=0$ into their correct equation
		$(-10, 0)$		A1	SC B2 for an answer of (18,0) or (0.5,0) oe or (7.5,0) oe
					Total 4 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 15

15	(Gradient of L_1 =) 6 ÷ 2 (=3)		4	M1	could be seen as part of an equation. Ignore constant term if candidate rearranges L_1
	$m \times$ "3" = −1 **or** $m = -\frac{1}{\text{"3"}}$			M1	for use of $m_1m_2 = -1$ could be seen as part of an equation
	$-1 = \text{"}-\frac{1}{3}\text{"} \times 9 + c$ **or** $y - -1 = \text{"}-\frac{1}{3}\text{"}(x-9)$ **or** $c = 2$			M1	
		$y + \frac{1}{3}x = 2$		A1	oe in required form eg $3y + x = 6$, $6y + 2x = 12$ etc
					Total 4 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 17

Question	Working	Answer	Mark	Notes
17	eg $\frac{4-(-1)}{6-4}$ $(= \frac{5}{2} = 2.5)$		4	M1 for a method to find the gradient of **L**
	eg $\frac{-1}{\text{"2.5"}}$ $(= -\frac{2}{5} = -0.4)$ or $\frac{-1}{\textit{their}\ \text{gradient}}$ oe			M1 ft for a method to find the gradient of **M** if *their* gradient of **L** clearly stated (even if no method shown for gradient of **L**)
	$y =$ "-0.4"$x + 8$ oe eg $y - 8 = -\frac{2}{5}(x-0)$ **or** $(8 \div 2) \times 5$ (= 20) oe **or** $8 \div$ (−'*their gradient of* **M**')			M1 dep on previous M1 for substitution of (0, 8) into equation for a line **or** use of $(8 \div 2) \times 5$ (= 20) (maybe on diagram) NB: 20 gains M3 if clearly intended as x coordinate (stated or on a diagram)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	(20, 0)		A1
				Total 4 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 18

18	$y = -2x(+9)$ or gradient of line = −2		4	M1	a rearrangement with the correct x term or stating that the gradient of given line is −2
	Gradient of perpendicular = $\frac{1}{2}$ oe eg $\frac{-1}{-2}$			M1ft	For a statement that the gradient of the perpendicular line is $\frac{1}{2}$ or implication by equation of line with gradient $\frac{1}{2}$ (if a student goes straight to this stage then M2 is awarded) **This mark can also be awarded for the perpendicular gradient of what they indicate the gradient of the original line to be**
	eg $11 = \text{"}\frac{1}{2}\text{"} \times 8 + c$ or $y - 11 = \text{"}\frac{1}{2}\text{"}(x-8)$ oe or $c = 7$			M1dep	dep on previous M1 being awarded; a correct method to find the equation of the perpendicular line by using their gradient of the perpendicular line and (8, 11)
	Correct answer scores full marks (unless from obvious incorrect working)	$y = \frac{1}{2}x + 7$		A1oe	a correct equation for the line in the form $y = mx + c$ (as requested) If no other marks awarded then award SCB1 for $y = -\frac{1}{2}x + 15$
					Total 4 marks

CHALLENGE QUESTIONS

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2019 · Question 21*

21	$y=\frac{6}{4}x(+33)$ **or** (gradient =) $\frac{6}{4}$ oe $m\times\frac{6}{4}=-1$ **or** (gradient of **M** =) $-\frac{2}{3}$ oe $\frac{k-6}{-4-5}=\text{"}-\frac{2}{3}\text{"}$	12	4	M1 M1 M1 dep A1	or complete method to find equation of line ($3y=-2x+28$) and then substitution of $x=-4$

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 19*

19 (a)		$y=-4x+k$ (oe)	1	B1	for $y=-4x$ or $y=-4x+k$ where k is any numerical value $k\neq 7$ Could be written in another form e.g. $3y+12x=20$
(b)	$m=\frac{-2-1}{2--3}$ *or* $m=\frac{1--2}{-3-2}$ or $-\frac{3}{5}$ or -0.6		4	M1	for using $m=\frac{y_2-y_1}{x_2-x_1}$
	$m_p=\frac{5}{3}$			M1ft	for using $m_1\times m_2=-1$
	$4=\frac{5}{3}(-6)+c$ oe eg $4=-10+c$ ($c=14$) $y-4=\frac{5}{3}(x--6)$			M1ft	dep on previous M1 for substituting $(-6, 4)$ into linear equation formula $4=\frac{5}{3}x+c$ to find value of c or $y=\frac{5}{3}x+14$ or $y=1.66...x+14$
		Eg $5x-3y+42=0$		A1	for correct simplified equation where all values are integers $10x-6y+84=0$ or $3y=5x+42$ oe
					Total 5 marks

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 25*

Q	Working	Answer	Mark	Notes
	$\left(\frac{4+8}{2},\frac{7-5}{2}\right)$ oe or (6, 1)		5	M1 for finding the midpoint of PR
	$\frac{7--5}{4-8}\left(=-\frac{12}{4}=-3\right)$ oe			M1 for method to find the gradient of PR
	$\text{"}-3\text{"}\times m=-1$ oe or $(m=)\frac{-1}{\text{"}-3\text{"}}$ or $(m=)\frac{1}{3}$			M1ft for finding the gradient of QS, may be seen embedded in an equation, ft their gradient of PR
	$\text{"}1\text{"}=\text{"}\frac{1}{3}\text{"}(\text{"}6\text{"})+c$ oe or $c=-1$ **or** $y-1=\frac{1}{3}(x-6)$ **or** $y=\frac{1}{3}x-1$			M1ft (dep on previous M1) for finding the equation through QS, ft their gradient of PR and their midpoint of PR, do not allow (4, 7) or (8, −5) as midpoint PR
	Correct answer scores full marks (unless from obvious incorrect working)	$3y=x-3$		A1 oe eg $6y=2x-6$ or $3y-x+3=0$ etc but must be integer coefficients accept $a=3$, $b=1$, $c=-3$
				Total 5 marks

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 23

23	$\left(\frac{-1+5}{2}, \frac{6-4}{2}\right)$ or $\left(\frac{4}{2}, \frac{2}{2}\right)$ or (2, 1)		6	M1 for finding midpoint
	$\frac{-4-6}{5--1}$ or $\frac{6--4}{-1-5}$ or $-\frac{10}{6}$ or $-\frac{5}{3}$			M1 indep for finding the gradient of PQ
	$\frac{-1}{-\frac{10}{6}}$ or $\frac{6}{10}$ or $\frac{-1}{-\frac{5}{3}}$ or $\frac{3}{5}$ or 0.6			M1 for finding the perpendicular gradient to PQ (ft their stated gradient)
	$1=\frac{3}{5}(2)+c$ or $c=-\frac{1}{5}$ or $c=-\frac{2}{10}$ or $c=-0.2$			M1 dep on 1st and 3rd M1 for substituting '(2, 1)' into $y='\frac{3}{5}'x+c$ or find the value of c oe eg $y-'1'='\frac{3}{5}'(x-'2')$
	$y=\frac{3}{5}x-\frac{1}{5}$ or $y=0.6x-0.2$ or $5y=3x-1$			A1 for a correct equation in any form
		$3x-5y-1=0$		A1 for $3x-5y-1=0$ or $5y-3x+1=0$ or $6x-10y-2=0$ oe accept in the form $ax+by=-c$ eg $3x-5y=1$ or $5y-3x=-1$ oe
				Total 6 marks

Alternative Mark Scheme for Q23

23	$(x+1)^2+(y-6)^2$ or $(x-5)^2+(y+4)^2$		6	M1
	$(x+1)^2+(y-6)^2=(x-5)^2+(y+4)^2$			M1 using $PA^2=QA^2$ (for some point A on the line)
	$x^2+2x+1+y^2-12y+36$ or $x^2-10x+25+y^2+8y+16$			M1
	$x^2+2x+1+y^2-12y+36=x^2-10x+25+y^2+8y+16$			M1
	eg $2x+1-12y+36=-10x+25+8y+16$ or $12x+37=20y+41$			A1 for a correct linear equation in x and y
		$3x-5y-1=0$		A1 for $3x-5y-1=0$ oe
				Total 6 marks

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Question 22 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 20*

20	$\left(\frac{-6+5}{2}, \frac{2+3}{2}\right) = \left(-\frac{1}{2}, \frac{5}{2}\right)$ oe		7	M1 for finding the midpoint of AB
	$\frac{2-3}{-6-5}\left(=\frac{-1}{-11}=\frac{1}{11}\right)$ oe			M1 for finding the gradient of AB
	$\frac{1}{11} = -1$ **or** $(m=)-11$			M1ft their gradient of AB (indep) for the correct use of $m_1 \times m_2 = -1$
	$"\frac{5}{2}" = "-11"\left("-\frac{1}{2}"\right) + c$ oe **or** $y - "\frac{5}{2}" = "-11"\left(x - "-\frac{1}{2}"\right)$ **and** $(y=)"-11"(-1) - 3 (=8)$ **or** $(y=)"-11"\left(-1 - "-\frac{1}{2}"\right) + "\frac{5}{2}" (=8)$			M1 for an expression that gives the y value at C
See alt methods	(Perp =) $\sqrt{\left(8-\frac{5}{2}\right)^2 + \left(-1--\frac{1}{2}\right)^2}\left(=\frac{\sqrt{122}}{2}\right)$ **and** $(AB=)\sqrt{3-2^{\,2}+\,5--6^{\,2}}\ (=\sqrt{122})$			M1
	(Area of triangle =) $\frac{1}{2} \times \sqrt{122} \times \frac{\sqrt{122}}{2}$			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)	30.5		A1 oe Allow answers in the range 30.4 – 30.5
				Total 7 marks

Q20 Alternative ways of finding the area of the triangle (final 2 M marks)				
Alt1	(11 × 6) – (0.5 × 1 × 11) – (0.5 × 5 × 6) – (0.5 × 5 × 6)			M1 for any 3 correct triangles
	(11 × 6) – (0.5 × 1 × 11) – (0.5 × 5 × 6) – (0.5 × 5 × 6)			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)			
Alt 2	$(AC = BC =)\sqrt{5^2+6^2}\ (=\sqrt{61})$			M1 for AC is perp to BC
	(Area of triangle =) $\frac{1}{2} \times \sqrt{61} \times \sqrt{61}$			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)			
Alt 3	$\sqrt{\left(8-\frac{5}{2}\right)^2 + \left(-1--\frac{1}{2}\right)^2}\left(=\frac{\sqrt{122}}{2}\right)$ **and** $(AM=)\sqrt{\left(-6--\frac{1}{2}\right)^2 + \left(2-\frac{5}{2}\right)^2}\left(=\frac{\sqrt{122}}{2}\right)$ or $(BM=)\sqrt{\left(5--\frac{1}{2}\right)^2 + \left(3-\frac{5}{2}\right)^2}\left(=\frac{\sqrt{122}}{2}\right)$			M1 for the height of the triangle, AM and BM where M is the midpoint of AB
	(Area of triangle =) $2 \times \frac{1}{2} \times \frac{\sqrt{122}}{2} \times \frac{\sqrt{122}}{2}$			M1 for a complete method
	Correct answer scores full marks (unless from obvious incorrect working)			
Alt 4	$(AC = BC =)\sqrt{5^2+6^2}\ (=\sqrt{61})$ and $(AB=)\sqrt{3-2^{\,2}+\,5--6^{\,2}}\ (=\sqrt{122})$			M1 for finding AC, BC and AB
	$\sqrt{\left(\frac{\sqrt{122}+2\sqrt{61}}{2}\right)\left(\frac{\sqrt{122}+2\sqrt{61}}{2}-\sqrt{122}\right)\left(\frac{\sqrt{122}+2\sqrt{61}}{2}-\sqrt{61}\right)\left(\frac{\sqrt{122}+2\sqrt{61}}{2}-\sqrt{61}\right)}$			M1 for applying Heron's formula
	Correct answer scores full marks (unless from obvious incorrect working)			